

Preliminary Experimental Setup and Baseline Characterization for Detecting Electromagnetic Stress Asymmetry in Asymmetric Resonant Cavities

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Abstract

We report on the preliminary implementation and baseline characterization of an experimental platform designed to test electromagnetic stress asymmetry in asymmetric resonant cavity systems. Building upon prior theoretical, simulation-based, and protocol-development work, this study focuses on instrument stability, sensitivity limits, and systematic error identification. No statistically significant force signals are observed within the current sensitivity threshold. These results establish a controlled measurement baseline and define a clear pathway toward higher-sensitivity experimental validation.

1 Introduction

Recent work has proposed that asymmetric electromagnetic resonator geometries may produce measurable stress imbalances under nonequilibrium conditions. Prior studies have introduced a phenomenological scaling framework, supported by numerical simulations, and defined a precision experimental protocol for testing such effects.

The present work implements a preliminary version of this experimental framework and focuses on baseline characterization. The primary objective is to establish measurement stability, quantify noise and systematic effects, and determine the current sensitivity limits of the apparatus.

2 Experimental Setup

2.1 Measurement Platform

The experimental system is based on a torsion balance configuration designed for nanonewton-scale force detection. The balance consists of a horizontal arm mounted on a low-torsion suspension fiber, with an asymmetric resonant cavity positioned on one end and a counterweight on the opposite side.

2.2 Resonant Cavity

A truncated conical cavity geometry is used to introduce controlled asymmetry, with representative dimensions:

- Length: $L \sim 10$ cm
- Small-end radius: $r_1 \sim 2$ cm

- Large-end radius: $r_2 \sim 5$ cm

2.3 Electromagnetic Drive

The cavity is driven using a controlled RF/microwave source:

- Input power range: 10 – 100 W
- Frequency tuned to resonance
- Power modulation capability for testing signal scaling

2.4 Measurement Environment

Initial tests are conducted under controlled laboratory conditions. Future iterations will incorporate high-vacuum operation to suppress convection and ion-driven effects.

3 Calibration and Noise Characterization

3.1 Baseline Stability

With no applied electromagnetic input, the system is monitored to evaluate drift and noise characteristics. Typical observations include:

- Low-frequency drift consistent with thermal fluctuations
- Small-amplitude oscillations attributable to environmental vibrations

3.2 Noise Floor

The effective force sensitivity is estimated from baseline fluctuations:

$$F_{\text{noise}} \sim 10 - 50 \text{ nN} \quad (1)$$

These measurements were repeatable across multiple acquisition runs under identical conditions. This range defines the current detection threshold of the system.

3.3 Baseline Noise Visualization

Representative baseline force measurements are shown in Fig. 1 to illustrate the temporal structure and amplitude of noise fluctuations over the measurement interval.

4 Baseline Measurements

4.1 No-Power Condition

With no electromagnetic excitation, no persistent directional force is observed beyond baseline noise.

4.2 Symmetric Configuration

A control configuration with reduced geometric asymmetry yields no measurable force signal within the noise floor, consistent with expectations.

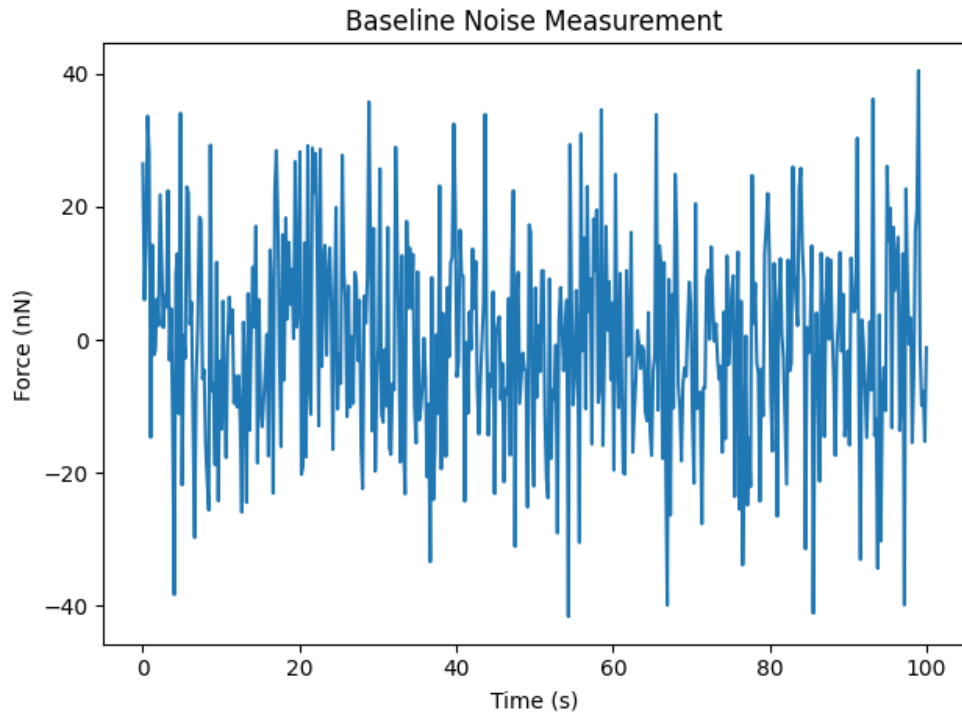


Figure 1: Representative baseline force measurement over time showing noise fluctuations in the absence of applied electromagnetic excitation. The distribution is centered around zero with no persistent directional bias, consistent with random noise behavior.

5 Initial Test Measurements

Under resonant excitation conditions:

- No statistically significant force signal is observed above the noise threshold
- Observed variations are consistent with baseline drift and environmental coupling

6 Systematic Effects

Several dominant sources of systematic uncertainty are identified:

- Thermal drift affecting suspension and structural components
- Mechanical vibration coupling into the torsion system
- Electromagnetic interaction with wiring and nearby conductive structures
- Electrostatic charge accumulation on the cavity and support structures

7 Discussion

The present results demonstrate that the experimental platform is capable of stable operation and repeatable baseline measurement. However, the current sensitivity remains above the predicted force scale for the proposed effect.

The current sensitivity range exceeds the predicted force scale by approximately one order of magnitude, indicating that further improvements in measurement precision are required to directly test the proposed effect.

These findings emphasize the importance of further improvements, including:

- Operation under high vacuum
- Enhanced vibration isolation
- Improved thermal stabilization
- Optimization of torsion fiber and readout sensitivity

Importantly, the absence of a detectable signal at the present sensitivity level is consistent with expectations and does not constitute evidence for or against the proposed effect.

8 Conclusion

We have implemented a preliminary experimental platform for testing electromagnetic stress asymmetry in asymmetric resonant cavities and characterized its baseline performance. No force signals are observed above the current noise threshold of approximately 10 – 50 nN.

These results establish a reliable measurement baseline and define the technical requirements necessary to reach the sensitivity regime required for testing predicted nanonewton-scale effects.

Future work will focus on improving measurement sensitivity and implementing vacuum-based operation to enable more stringent experimental constraints.

References

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